Lab2: Basics of Computational Programming and Software

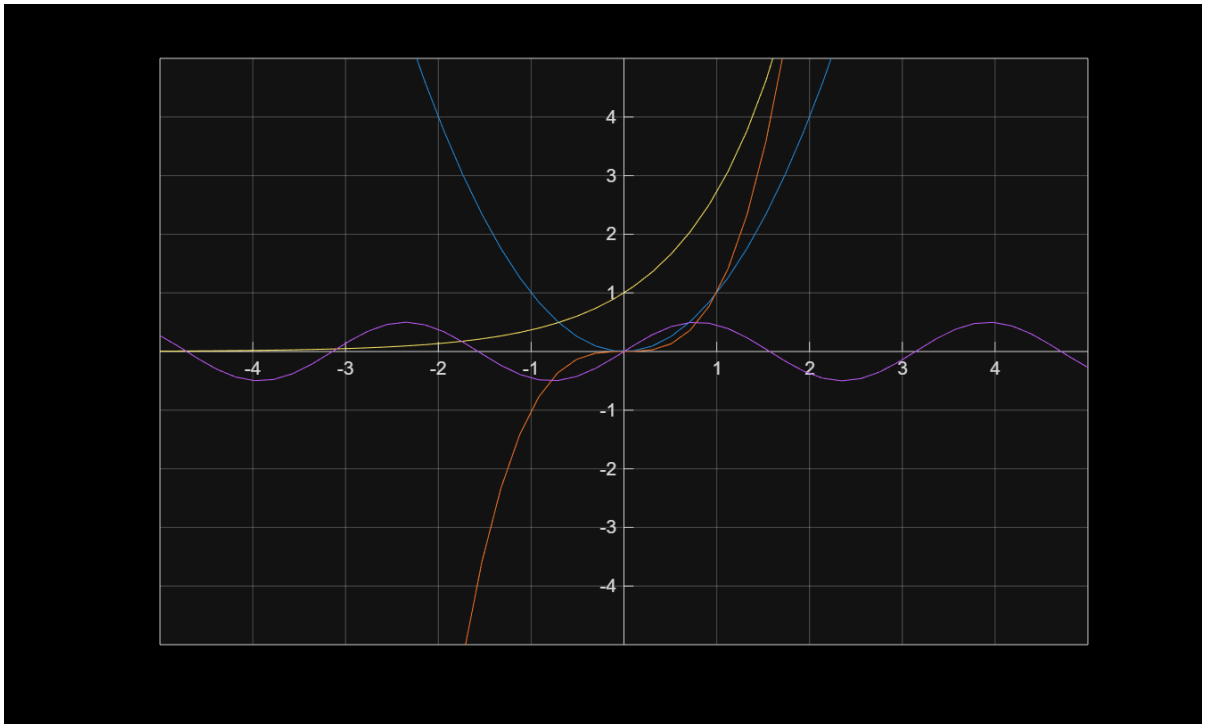
Question 1:

Draw the functions

$$y_1 = x^2$$
$$y_2 = x^3$$
$$y_3 = e^x$$
$$y_4 = \sin(x) * \cos(x)$$

on same plot for the range of x and y from -5 to +5. Also label the plots aesthetically.

```
clc;
clear;
x1 = linspace(-5,5,50);
y1 = x1.^2;

y2 = x1.^3;

y3 = exp(x1);

y4 = sin(x1).*cos(x1);

plot(x1,y1,x1,y2,x1,y3,x1,y4)
xlim([-5,+5]);
ylim([-5,+5]);
grid on;
axis on;

ax = gca;
ax.XAxisLocation = 'origin';
ax.YAxisLocation = 'origin';

saveas(gcf, "Exp2_Q1.png")
```



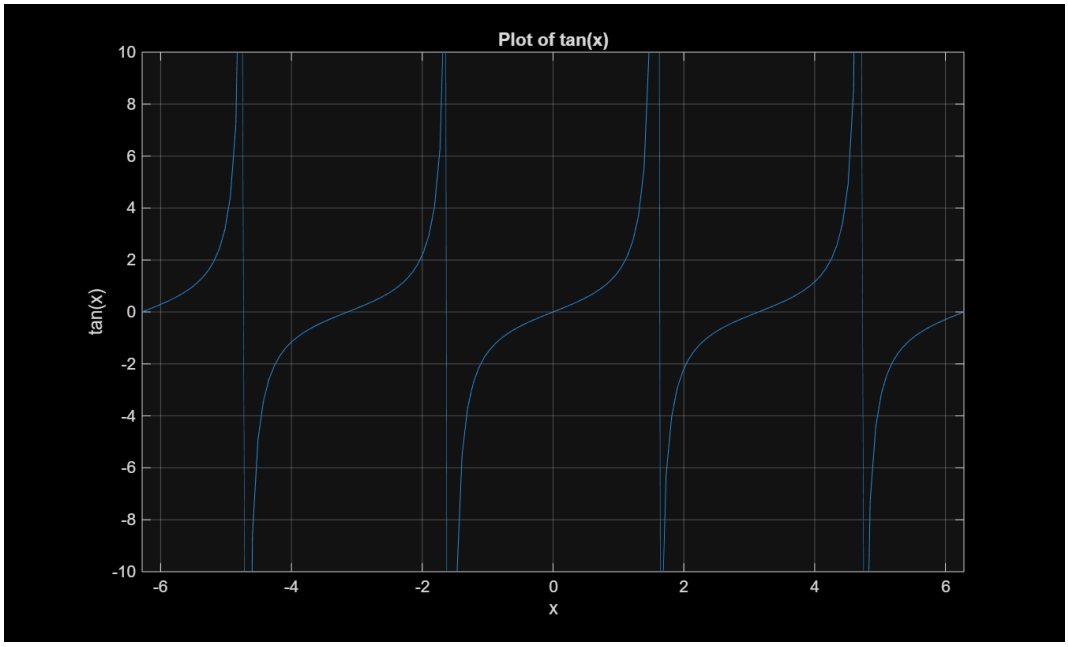
Question 2:

Make a MATLAB program that sums number from 1 to 100 using FOR loop statement.

```
clc;clear;
sum = 0;
for i = 1:1:100
    sum = sum + i;
end

disp("The sum of numbers ranging 1 to 100 is:")
disp(sum
```

```
Command Window
The sum of numbers ranging 1 to 100 is:
        5050

>> Press Ctrl + Shift + P to generate code with Copilot
```

Question 3:

Write a MATLAB code to plot 'tan' function with x domain as -2π to 2π with 150 number of points. (Hint – when you plot, you will notice the plot is not aesthetic. To make your plot aesthetic, use – xlim and ylim functions. Example - xlim([-2*pi 2*pi]); ylim([-10 10]);

```
x = linspace(-2*pi,2*pi,150);
y = tan(x);

plot(x, y);
xlim([-2*pi, 2*pi]);
ylim([-10, 10]);
xlabel('x');
ylabel('tan(x)');
title('Plot of tan(x)');
grid on;
saveas(gcf, "Exp2_Q3.png")
```